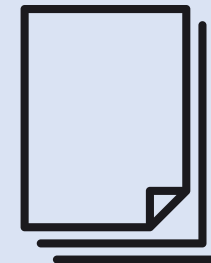
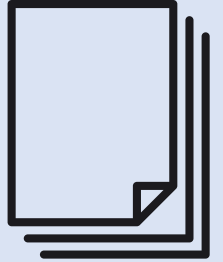


Starter



1. Find $\frac{d}{dx}(xy)$
2. Find $\frac{d}{dx}(y \sin x)$
3. Find $\frac{d}{dx}(xy^2)$
4. Find $\frac{d}{dx}(\sinh(x)e^y)$
5. Find $\frac{d}{dx}(\ln(x) \tan(2y))$

Starter



1. Find $\frac{d}{dx}(xy) = x \frac{dy}{dx} + y$

2. Find $\frac{d}{dx}(y \sin x) = y \cos x + \sin x \frac{dy}{dx}$

3. Find $\frac{d}{dx}(xy^2) = y^2 + 2xy \frac{dy}{dx}$

4. Find $\frac{d}{dx}(\sinh(x)e^y) = \cosh(x)e^y + \sinh(x)e^y \frac{dy}{dx}$

5. Find $\frac{d}{dx}(\ln(x) \tan(2y)) = \frac{\tan(2y)}{x} + 2 \ln(x) \sec^2(2y) \frac{dy}{dx}$

What is a differential equation?

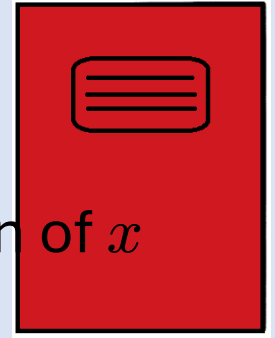
Technically, this is an equation involving a variable, x , a function of x (usually ' y '), and the derivative of the function $\frac{dy}{dx}$.

The goal is to work out what function satisfies that equation.

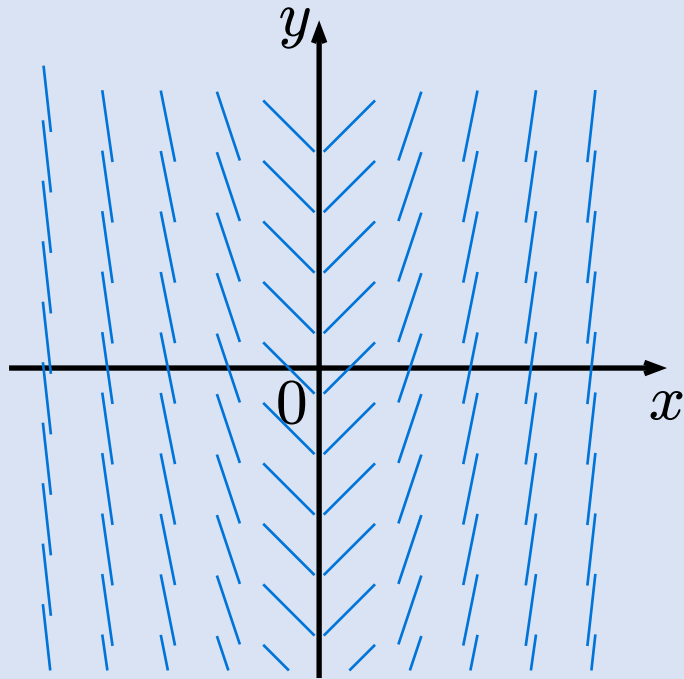
It's often easiest to imagine these as a **vector field**.

EG.

$$\frac{dy}{dx} = 2x$$

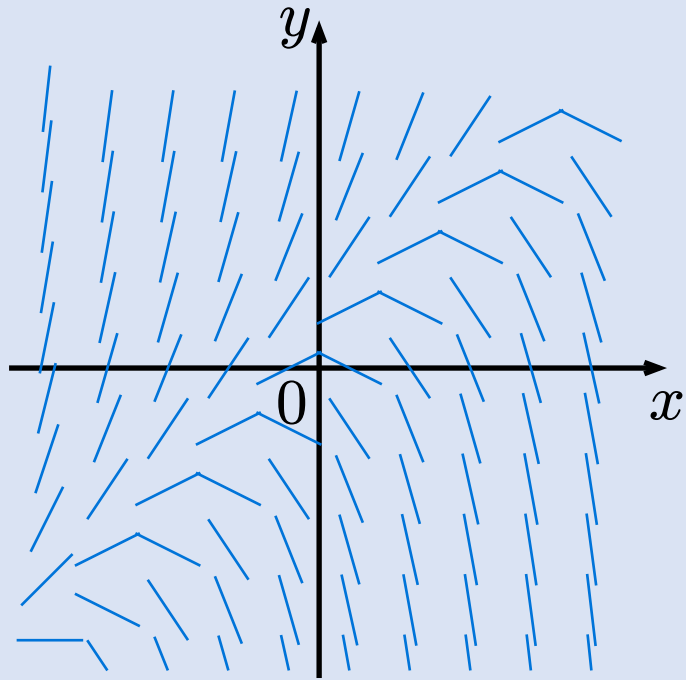


$$\frac{dy}{dx} = 2x$$



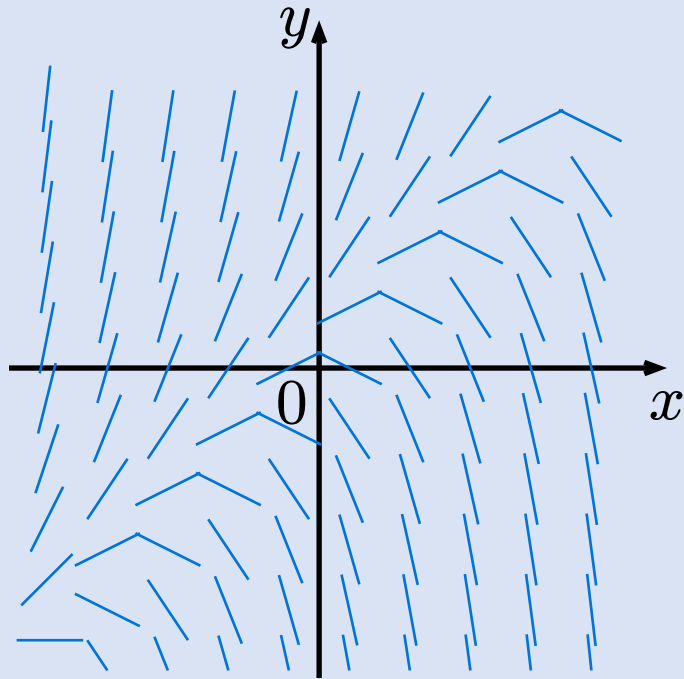
The gradient at any point on this graph can be calculated as $2x$, so we'll see a quadratic if we choose any specific point.

$$\frac{dy}{dx} = y - x$$



Every point within this space has a direction and a magnitude.

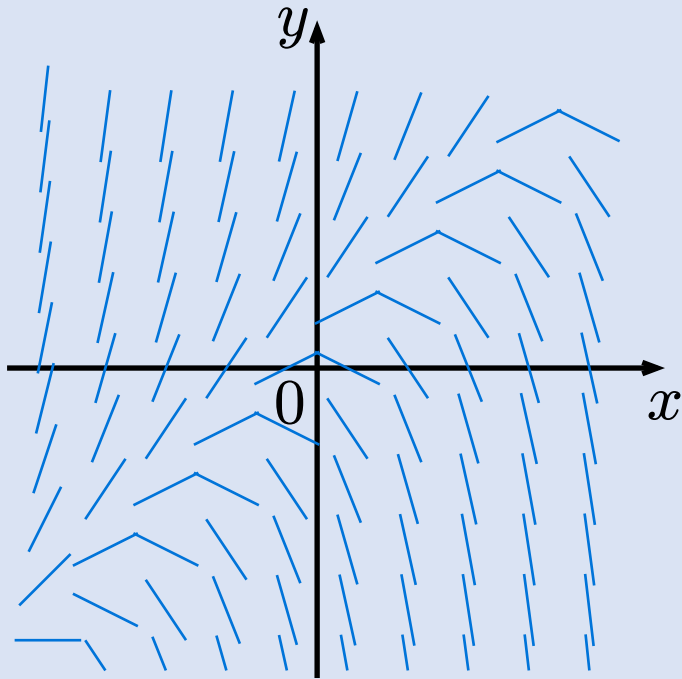
$$\frac{dy}{dx} = y - x$$



Every point within this space has a direction and a magnitude.

If we want to find a particular function in here we can imagine dropping a point into this field and watching where it goes.

$$\frac{dy}{dx} = y - x$$



Every point within this space has a direction and a magnitude.

If we want to find a particular function in here we can imagine dropping a point into this field and watching where it goes.

More formally, this is **integration**.
What does the $+c$ represent then?

Differential Equations

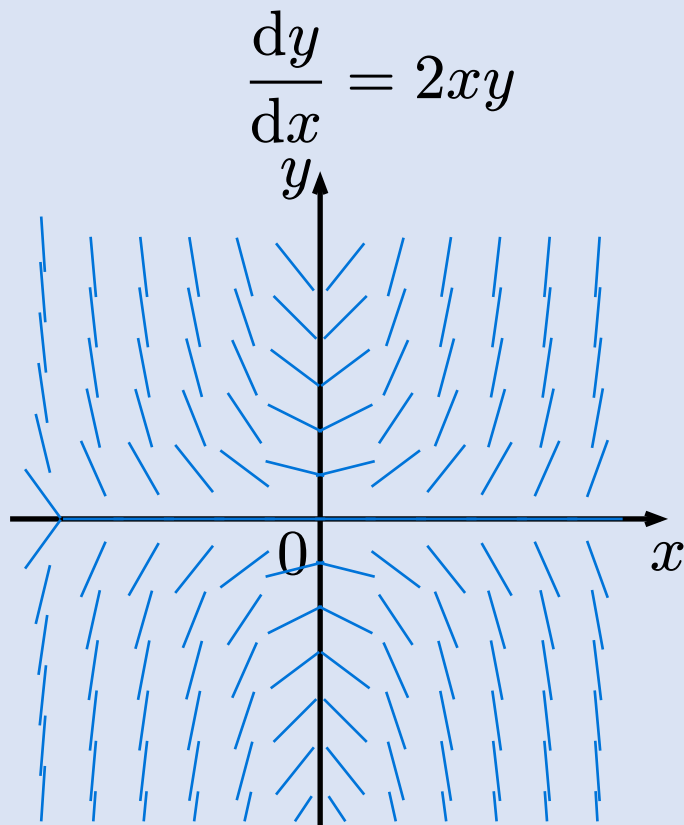
What kinds of shapes do these more complicated equations make?

$$\frac{dy}{dx} = 2xy$$

$$\frac{dy}{dx} = 3x + y$$

Differential Equations

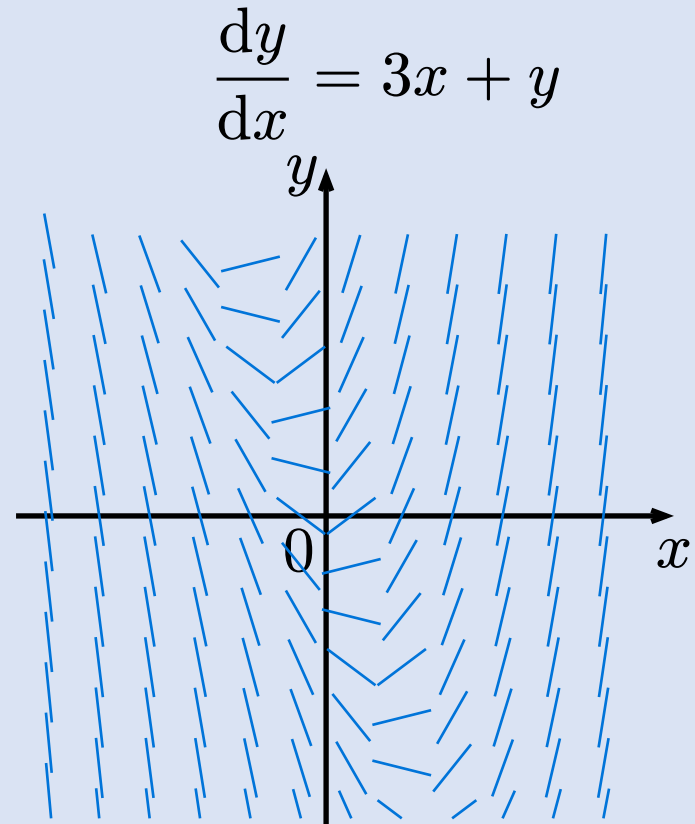
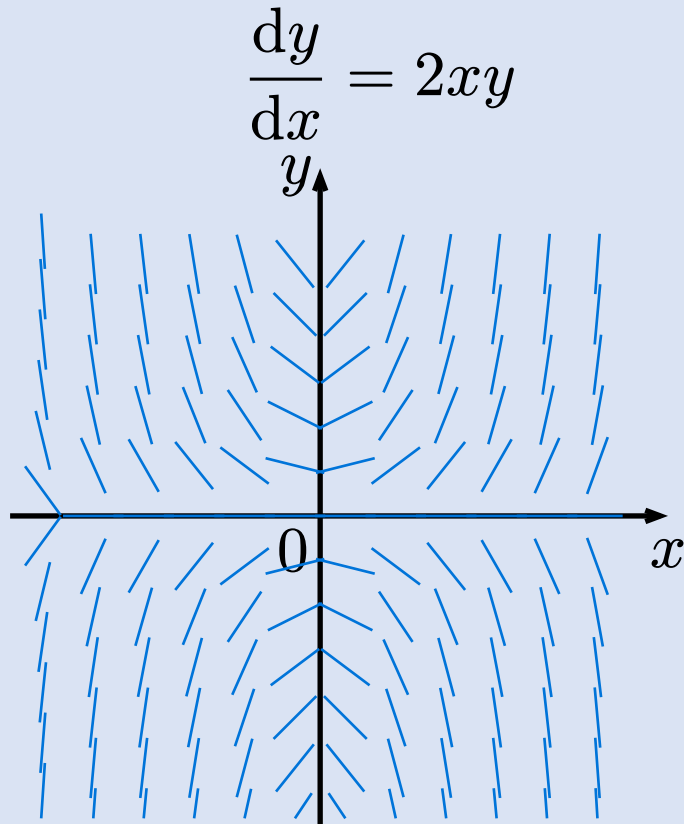
What kinds of shapes do these more complicated equations make?



$$\frac{dy}{dx} = 3x + y$$

Differential Equations

What kinds of shapes do these more complicated equations make?



Differential Equations

Sometimes they are more complicated.

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = e^x$$

This means that there is some function y such that the sum of y , the first derivative and the second derivative makes e^x

Differential Equations

Sometimes they are more complicated.

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = e^x$$

This means that there is some function y such that the sum of y , the first derivative and the second derivative makes e^x

We'll look at techniques to solve a small number of different types of DEs on this course, but they do go much further.

So far in A Level Maths, you've used **Separation of Variables**

Drawing Fields



For $-4 \leq x \leq 4$ and $-4 \leq y \leq 4$ sketch the family of solutions for

Drawing Fields



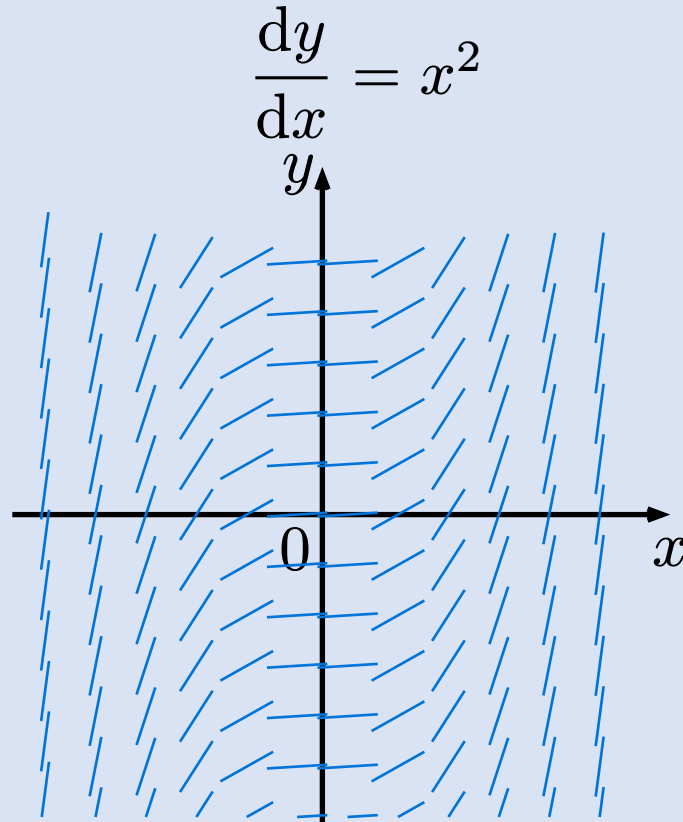
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Drawing Fields



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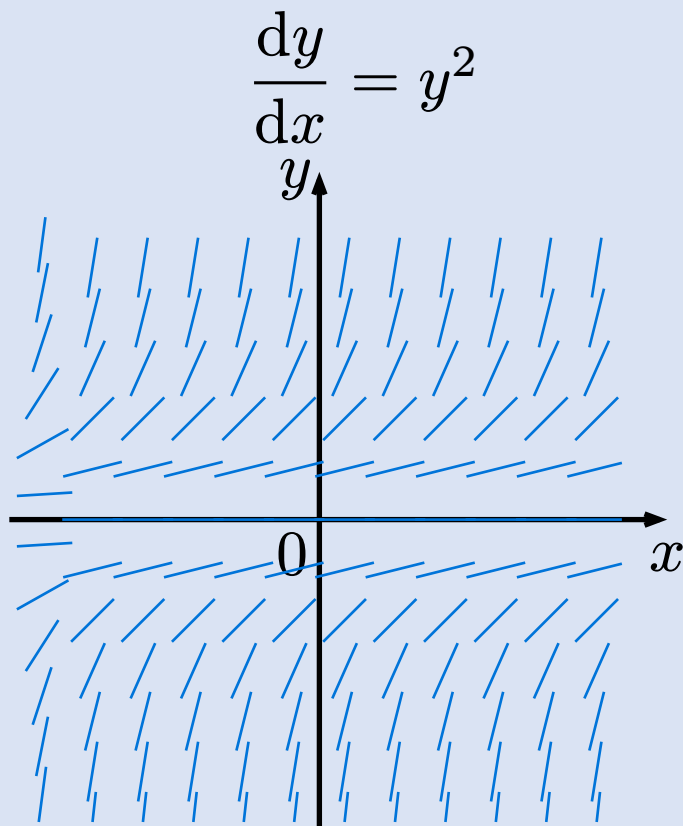
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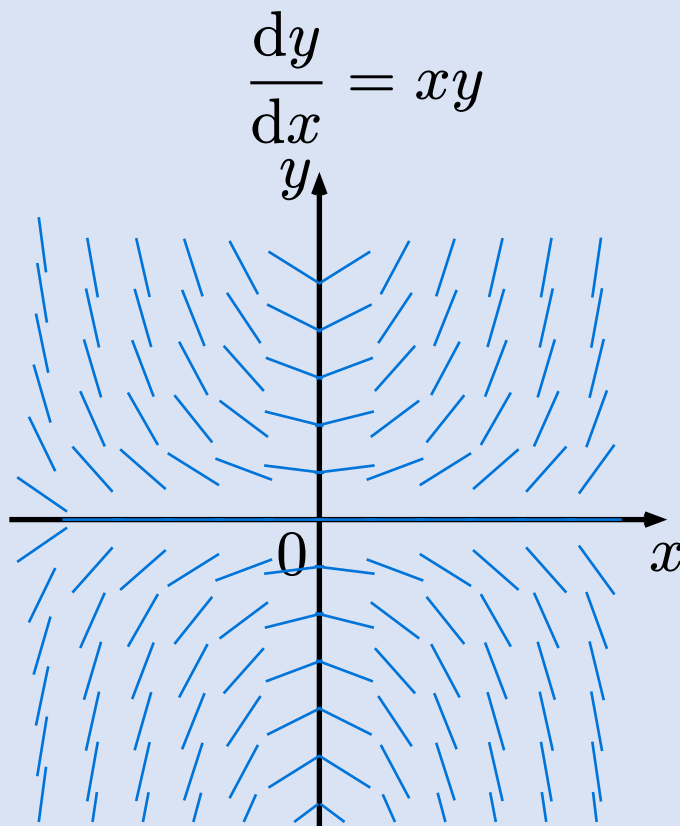
For $-4 \leq x \leq 4$ and $-4 \leq y \leq 4$ sketch the family of solutions for

$$\frac{dy}{dx} = xy$$

Drawing Fields



For $-4 \leq x \leq 4$ and $-4 \leq y \leq 4$ sketch the family of solutions for



Drawing Fields



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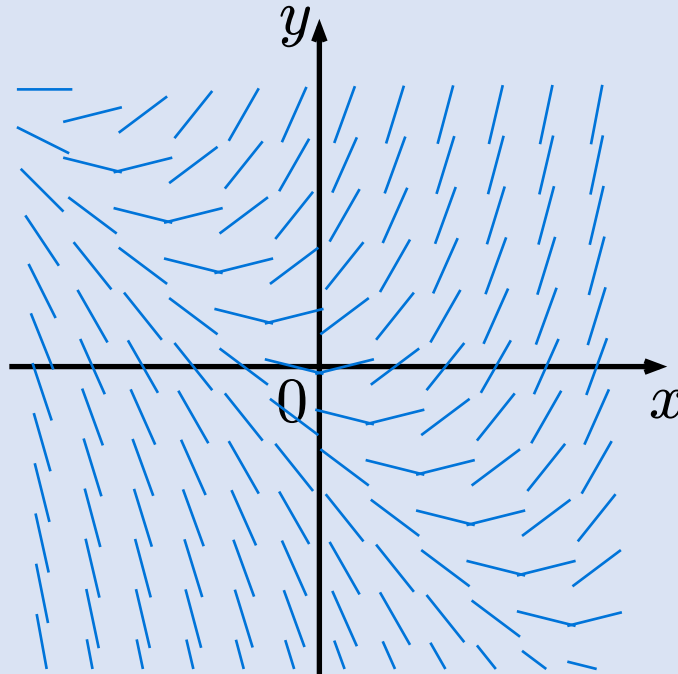
$$\frac{dy}{dx} = x + y$$

Drawing Fields



For $-4 \leq x \leq 4$ and $-4 \leq y \leq 4$ sketch the family of solutions for

$$\frac{dy}{dx} = x + y$$



Drawing Fields



For $-4 \leq x \leq 4$ and $-4 \leq y \leq 4$ sketch the family of solutions for

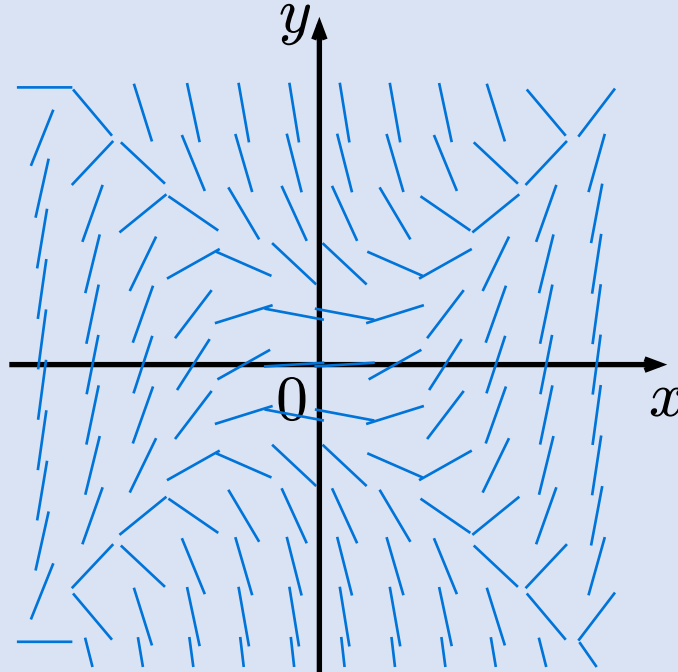
$$\frac{dy}{dx} = x^2 - y^2$$

Drawing Fields



For $-4 \leq x \leq 4$ and $-4 \leq y \leq 4$ sketch the family of solutions for

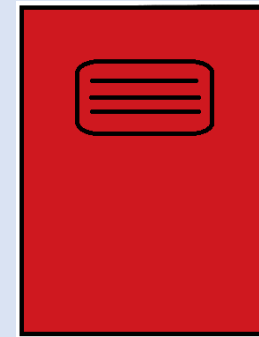
$$\frac{dy}{dx} = x^2 - y^2$$



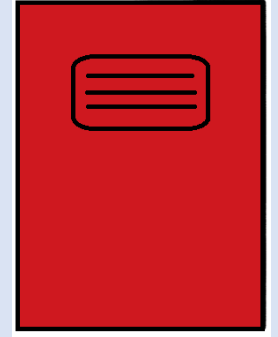
Separation of Variables - Recap

Find the general solution to

$$\frac{dy}{dx} = \frac{2x}{y}$$



Separation of Variables - Recap



Find the general solution to

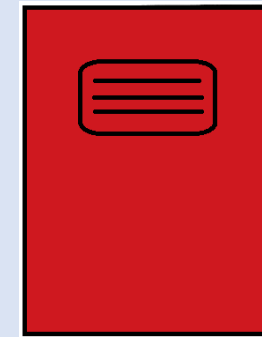
$$\frac{dy}{dx} = \frac{2x}{y}$$

$$y \frac{dy}{dx} = 2x \Rightarrow \int y \, dy = \int 2x \, dx$$

$$\therefore \frac{1}{2}y^2 = x^2 + c \Rightarrow x^2 - \frac{1}{2}y^2 + c = 0$$

Given that $y = 0$ when $x = 1$, find c .

Separation of Variables - Recap



Find the general solution to

$$\frac{dy}{dx} = \frac{2x}{y}$$

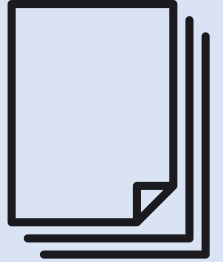
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$$\therefore \frac{1}{2}y^2 = x^2 + c \Rightarrow x^2 - \frac{1}{2}y^2 + c = 0$$

Given that $y = 0$ when $x = 1$, find c .

$$1 - 0 + c = 0 \Rightarrow c = -1 \Rightarrow x^2 - \frac{1}{2}y^2 - 1 = 0$$

Recap: You Try



Find the solution to each differential equation, given the initial conditions.

1. $\frac{dy}{dx} = xy, x = 0, y = 2$

2. $\frac{dy}{dx} = \frac{e^y}{3}, x = e, y = -1$

3. $\frac{dy}{dx} = \frac{\sin x}{y}, x = \frac{\pi}{2}, y = 1$

4. $\frac{dy}{dx} = \frac{1+y^2}{2}, x = 1, y = 0$

5. $\frac{dy}{dx} = y - x, x = 0, y = 3$

solutions

$$\frac{dy}{dx} = xy, x = 0, y = 2, x = e, y = -1$$

$$\int \frac{1}{y} dy = \int x dx$$

$$\ln y = \frac{1}{2}x^2 + c$$

$$y = e^{\frac{1}{2}x^2 + c} \equiv Ae^{\frac{1}{2}x^2}$$

$$y|_{(0,2)} = 2 = Ae^0 \Rightarrow A = 2$$

$$\therefore y = 2e^{\frac{1}{2}x^2}$$

solutions

$$\frac{dy}{dx} = \frac{e^y}{3}$$

$$\int e^{-y} dy = \int \frac{1}{3} dx$$

$$-e^{-y} = \frac{1}{3}x + c \Rightarrow y = -\ln\left|c - \frac{1}{3}x\right|$$

$$y|_{(e,-1)} = -1 = -\ln\left(c - \frac{1}{3}e\right)$$

$$e = c - \frac{1}{3}e \Rightarrow c = \frac{4}{3}e$$

$$\therefore y = -\ln\left(\frac{4}{3}e - \frac{1}{3}x\right)$$

solutions

$$\frac{dy}{dx} = \frac{\sin x}{y}, x = \frac{\pi}{2}, y = 1$$

$$\int y \, dy = \int \sin x \, dx \Rightarrow \frac{1}{2}y^2 = -\cos x + c$$

$$y^2 = c - 2\cos x \mid_{(\frac{\pi}{2}, 1)} = 1 = c - 2 \times 0 \Rightarrow c = 1$$

$$\therefore y^2 = 1 - 2\cos x \text{ or } y = \sqrt{1 - 2\cos x}$$

solutions

$$\frac{dy}{dx} = \frac{1+y^2}{2}, x = 1, y = 0$$

$$\int \frac{1}{1+y^2} dy = \int \frac{1}{2} dx \Rightarrow \arctan y = \frac{1}{2}x + c$$

$$@ (1, 0) = 0 = \frac{1}{2} + c \Rightarrow c = -\frac{1}{2}$$

$$\therefore \arctan y = \frac{1}{2}x - \frac{1}{2} \text{ or } y = \tan\left(\frac{1}{2}x - \frac{1}{2}\right)$$

solutions

$$\frac{dy}{dx} = y - x, x = 0, y = 3$$

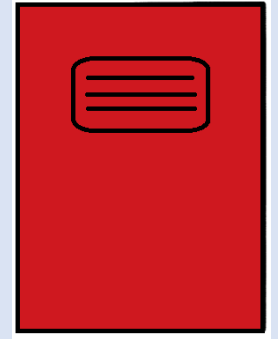
Can't be solved right now. We can't separate the variables.

Sorry!

We'll look at some options for these now and next lesson.

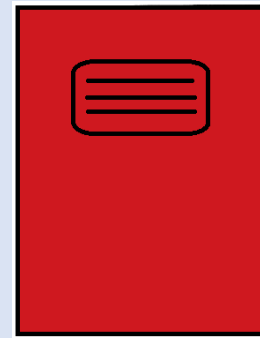
From the Starter

Find $\frac{d}{dx}(xy)$



From the Starter

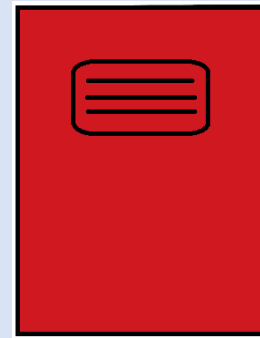
Find $\frac{d}{dx}(xy) = x \frac{dy}{dx} + y$



From the Starter

Find $\frac{d}{dx}(xy) = x \frac{dy}{dx} + y$

$$\therefore \int x \frac{dy}{dx} + y \, dx = xy + c$$



From the Starter

Find $\frac{d}{dx}(xy) = x \frac{dy}{dx} + y$

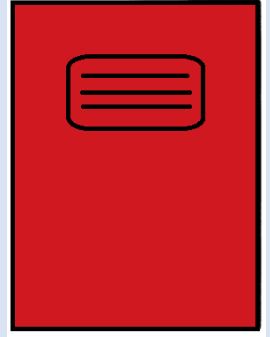
$$\therefore \int x \frac{dy}{dx} + y \, dx = xy + c$$

So if we had:

$$x \frac{dy}{dx} + y = x^2$$

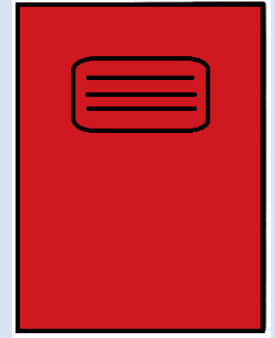
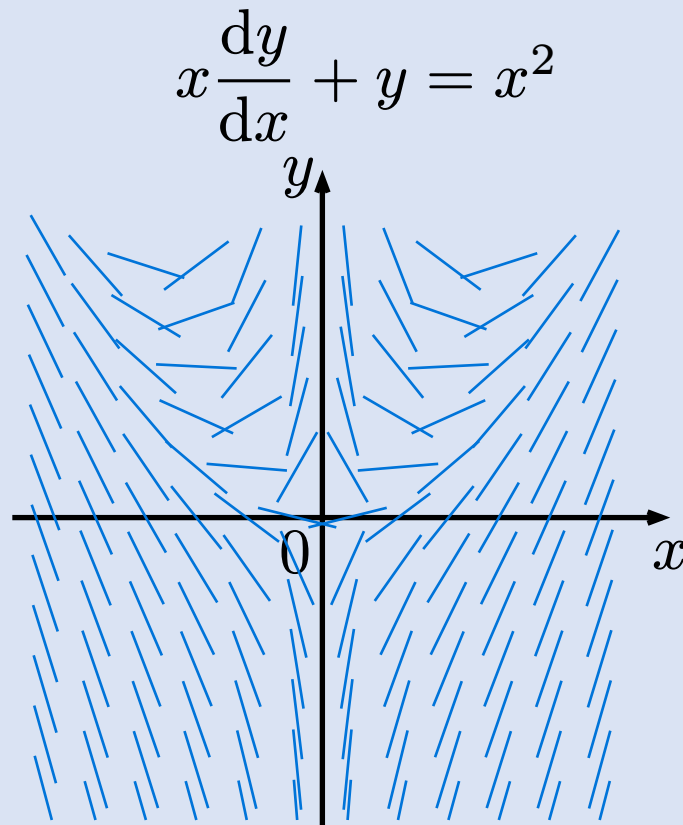
We could integrate both sides, and get:

$$\int x \frac{dy}{dx} + y \, dx = \int x^2 \Rightarrow xy = \frac{1}{3}x^3 + c \Rightarrow y = \frac{1}{3}x^2 + \frac{c}{x}$$



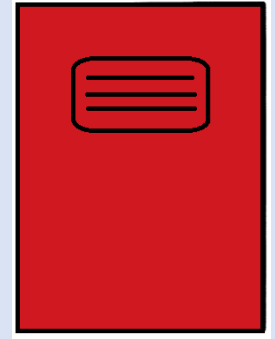
From the Starter

Find $\frac{d}{dx}(xy)$



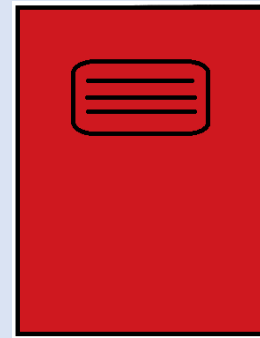
From the Starter

Find $\frac{d}{dx}(y \sin x)$



From the Starter

Find $\frac{d}{dx}(y \sin x) = y \cos x + \sin x \frac{dy}{dx}$

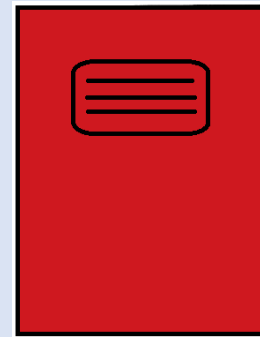


From the Starter

Find $\frac{d}{dx}(y \sin x) = y \cos x + \sin x \frac{dy}{dx}$

$$\therefore \int \frac{dy}{dx} \sin x + y \cos x \, dx = y \sin x + c$$

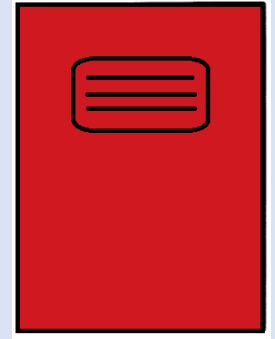
Which is effectively reversing the product rule!



Example

Find the General Solution to:

$$2xy \frac{dy}{dx} + y^2 = x^2$$



Example

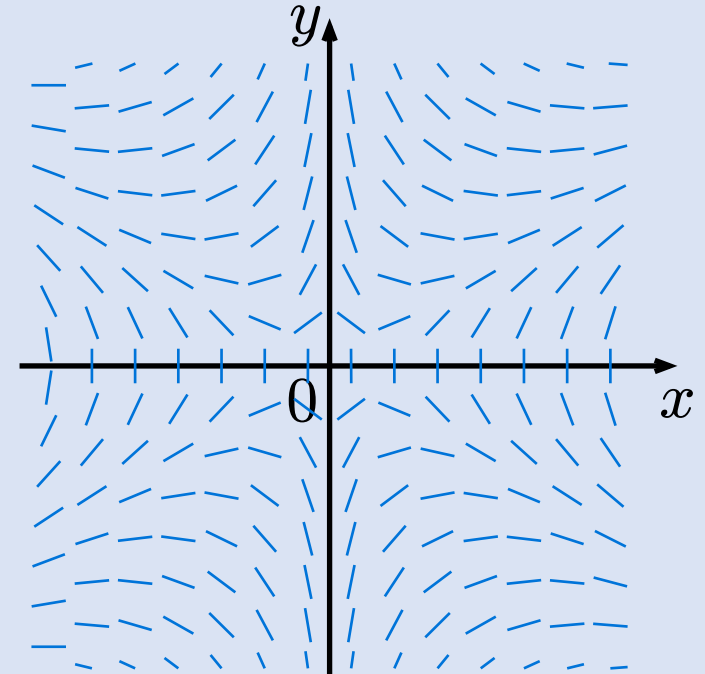
Find the General Solution to:

$$2xy \frac{dy}{dx} + y^2 = x^2$$

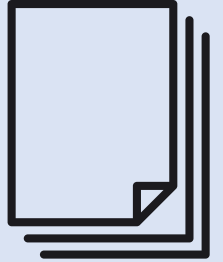
$$y^2 \rightarrow 2y \frac{dy}{dx}, x \rightarrow 1, \therefore \text{product rule!}$$

$$\int 2xy \frac{dy}{dx} + y^2 dx = \int x^2 dx$$

$$xy^2 = \frac{1}{3}x^3 + c \Rightarrow y^2 = \frac{1}{3}x^2 + \frac{c}{x}$$



Practice Time!



Find the general solutions by reversing the product rule:

1. $x^3 \frac{dy}{dx} + 3x^2 y = \frac{1}{x}$

2. $\sqrt{x} \frac{dy}{dx} + \frac{1}{2\sqrt{x}} y = x + 1$

3. $\cosh(x) \frac{dy}{dx} + \sinh(x) y = 2$

4. $\ln(x) e^y \frac{dy}{dx} + \frac{e^y}{x} = x$

5. $x^3 y^2 \frac{dy}{dx} + x^2 y^3 = 2$

6. $x^2 \frac{dy}{dx} + 4xy = 3$

Alternative Method

Use the substitution $z = y^2$ to find the general solution to

$$2xy \frac{dy}{dx} + y^2 = 2x + 1$$

Alternative Method

Use the substitution $z = y^2$ to find the general solution to

$$2xy \frac{dy}{dx} + y^2 = 2x + 1$$

$$z = y^2 \Rightarrow \frac{dz}{dx} = 2y \frac{dy}{dx}$$

$$\therefore x \frac{dz}{dx} + z = 2x + 1$$

$$xz = x^2 + x + c$$

$$z = x + 1 + \frac{c}{x} = y^2$$